

COMBINED SEWER OVERFLOWS IN BRAZIL: A 2014 SITUATION REPORT

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Abstract — Combined sewer systems are common at cold regions, where heavy storms, that disturb wastewater treatment processes by combined sewer overflows (CSO), are rare. However, in warm countries, the ageing process of collection pipes (causing underground leakage of rain water into separate sewers) and illicit connections of drainage pipes into sewers (increasingly common, either inside premises or between public, sewer and drainage systems) cause wastewater treatment plant inflow surges and quality disturbances to raw and treated sewage. Such perturbations bear close similarity to CSO impacts, and cause plants to yield low quality effluents – thus, jeopardizing receiving-water quality. The article briefly introduces the CSO problem in the world, first. Then, it reports the current situation at some Brazilian cities and towns that are fighting the problem by legal measures, and campaigns for educating citizens and companies such as to reduce illicit connections and preserve the integrity of these hydraulic systems.

Index Terms — Combined sewer overflow, storm effluent control, illicit sewer connection, dynamic plant operation, environmental education.

INTRODUCTION: FAST GROWTH AND SANITATION

Brazil started its European-influence development 514 years ago, but its more intense urbanisation began only in the late XIX Century, given the strong need for workers to staff agriculture – which was starting to flourish, then.

Southeast Brazil economic development was astonishing during the XX century. The case of São Paulo City industrial development is a particularly example of this. It was founded in 1554, at the neighbourhoods of a Christian Church, and school. It would officially know its population only at the demographic evaluation of 1872 – i.e., after 318 more years of history – when a counting of 20,000 souls was concluded. By influence of the coffee boom, it rose to 65,000 people in 1890 and to 270,000 in 1908. Then, it doubled to 578,000 in 1920 – after only 12 years. It had 1,120,000 people in 1934, so having faced a 5,500% growth in 62 years, only! Growth rates continued so astonishing until the late 1970s, allied to fast verticalisation [1] [2]. Meanwhile, some other big cities in Brazil were growing fast – though, perhaps, not so fast.

For the purpose of this study, it is important to note that, while São Paulo was growing so quickly, particularly in the 1920s, the lack of construction professionals was so severe

that 75% of the bricklayers had to come from Italy [2] – not to mention others, from Germany, Portugal and Spain. Consequently, most buildings were designed and raised only by bricklayers, electricians, plumbers or other practical workers, since the city graduated its first civil engineers only in 1898; and the first architects, in 1920 – even so, at numbers much lower than needed by that frenetic urban growth. However, the practical (and even technical) knowledge of several of these non-university-educated professionals had been developed still under the influence of a previously flourishing European culture – which gave them sufficient expertise to build in Brazil, at great numbers, houses that marked an era. Some of São Paulo traditional boroughs rose from the end of the XIX century to the beginning of last century, under the influence of the ‘coffee barons’ – rural landlords who developed the flourishing coffee agriculture that fuelled the prosperous ‘Paulista’ economy, responsible for the foundation of that industrial era. Examples of these neighbourhoods are Paulista Avenue and Higienópolis. There, some rare examples of neo-classic, two-store mansions escaped the skyscraper-building ‘fever’ that started at the middle of Century XX – and continues. Figure 1 is a rare example of their remainders.



FIGURE. 1
HOUSES FROM THE EARLY XX CENTURY IN ANGÉLICA AVENUE
(AT THE SQUARE OF HIGIENÓPOLIS AVENUE)

Only in 1933, at the foundation of the CONFEA/CREA system (federal/state boards, respectively) of professional boards, the rules were clearly set for professional exercise in Engineering, Architecture and Agronomy. Since then, such practice by unqualified personnel became irregular [2] [3]. However, the practice of construction by unauthorised workers persists today, particularly among the poorest sectors of society – in small towns or rural areas, and fast-growth communities or shantytowns in the metropolitan areas. All this helps to understand the lack of urban planning in many Brazilian cities and their sanitary problems, exemplified further here in the case of the Greater São Paulo.

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Following the facts reviewed above, a post-war, also fast urbanisation process happened from the middle of the XX Century, accompanied by the modernisation of the already implemented industry, spreading from São Paulo State onto most of the Southeast and South of Brazil. Thence, some larger cities begun to reach metropolitan sizes, as a relevant part of the rural population migrated into them. Much of this urban occupation by rural people occurred naturally, in a casual fashion – without a deeper thinking of public infrastructure, or services.

Fortunately, still before this period, Brazilian sanitary engineers stated a strong preference for separate sewers as the official type of wastewater collectors to be implemented nationwide. The conscience leading to this government directive was due to a crucial observation: an incipient, combined system installed in central São Paulo by foreigner contractors (late XIX Century, more precisely) was causing combined sewer overflows (CSO) [4]. Thanks to this, at that point – when São Paulo still was a relatively small City – its sanitary progress followed an appropriate standard in what concerns to the conception of wastewater collectors. Yet, despite such directives, which are still valid, flow surges at treatment-plant inlets have been observed in the São Paulo Metropolitan Area after rainfall events of high intensity on their respective collection areas [5] [6] [7], as well at smaller communities of more recent foundation in São Paulo State [8]. In addition, their frequency has intensified along the time and their flow rises are usually followed by important quality oscillations. At a first glance, this may seem odd, since this is a problem typical of another kind of sewers: combined sewer (called “sistema unitário”, in Brazil), usually built in colder, developed regions where intense, summer storms are rare. The reasons for such perturbations are seen further below. At this point, it is important to stress that they may cause relevant quality losses in the treated sewage; thus, similar impacts on the receiving water quality – particularly because a Brazilian WWTP is designed to treat separate sewage (or ‘dry-weather flow’ – DWF), mainly.

This paper reviews the incidence of CSO events in São Paulo State and other parts of Southeastern Brazil. However, it initially reviews the general aspects of this phenomenon, which have been disturbing wastewater treatment for decades at developed countries, adopting combined systems.

THE CSO PROBLEM WORLDWIDE

The earliest combined sewer systems of modern times were built in some European capital cities, particularly in England. A relevant growth in their numbers was observed from the middle of the XIX century. Following this, in the early XX century, these systems received many WWTP additions in order to give their sewage appropriate quality before its discharge into the aquatic environment. There were strong motivations, then, for collecting storm water and treating it together with sanitary sewage, the first one being: after a dry period, the first storm water flush tends to scour the pollution settled either on the ground surface or on sewer

bottom, conveying it downstream to receiving fluvial waters. Second, the manure disposed by traction animals over the roads (a reality until the early XX Century, at least) would increase the pollution load every time it rained and reach the river. Third, this organic load also increases as trees lose their leaves, especially during the autumn. Therefore, not treating storm waters was not an option for those societies, given their earnest plans to restore river water quality – majorly fulfilled, nowadays. Reda reviewed the origins of this fascinating subject, its consequences to wastewater treatment for century, and the solutions proposed to it are seen in [9].

Due to the influence of North European colonisation, but also to the presence of cold weather, the development of the North American eastern coast followed the same patterns of North Europe, mentioned above. Therefore, the CSO problem is common in the oldest urban communities of the USA (particularly in the North and Northeast) and Canada [10]. Aiming at curbing the impacts of this problem on river water quality, Beck proposes a dynamic approach denominated ‘Operational Water Quality Management’ [11], which amalgamates wastewater system operation and river quality management within a framework to be designed for mitigating CSO impacts onto the treatment process – and, consequently, onto receiving waters. That may also allow for further optimising the use of commonly finite resources for river water-quality conservation (or recovery). It combines real time simulation of system components (e.g., treatment plant; receiving water body; sewers hydraulic controls) to dynamic plant operation, as a methodology for the water industry to preserve river water quality in countries adopting combined collectors as a standard. In the case of developing countries with warm climates (i.e., subject to heavy rainfall), the challenge is represented by the increasing trend of illicit storm water intrusions into sewers due to fast urbanisation. This is the case of Brazil, where CSO surges may even increase (in frequency and intensity) due to climate changes.

In terms of river water-quality simulation under transient changes, the modelling framework presented in [9] can estimate flow and quality oscillations on the river for every chosen alternative of treatment strategy at a wastewater treatment plant (WWTP) discharging into it. The quality reached at the river may represent, in the near future, an objective function for environmental quality optimisation (see [12]). This framework includes treatment process simulation and water quality modelling on the receiving river, being applicable for system planning, design and operation – and, whenever opportune, eventually including real time sewer monitoring, simulation and control [13] [14] [15] [16]. One of the strongest conclusions of that research line was the important role of storm tanks in storing overflows at WWTP intakes [9] [10] – allowing for their treatment after the storm surge passes. Another strong conclusion was how important it is to grant treatment to the storm-inflow first flush and, as plant capacity is reached, deviate the excess to storm tanks. Such a control strategy was still novel at that time. However, on current days, its efficiency is well known. A contemporary

example of the application of these storm tanks is described in [17], on an advanced secondary treatment, economically succeeded, award-winning facility in Dallas with off-line storage of the wastewater excess. It beats the alternative of treating blended (waste-plus-rain) inflows at faster treatment speeds – not to mention the unsuitable alternative of increasing plant capacity only for the sake of CSO episodes.

THE OCCURRENCE OF CSO EPISODES IN BRAZIL

In Brazil, storm waters officially go to municipal drainage systems; then, to a surface water body. On the other hand, sewers conduct sanitary sewage, preferably into a WWTP. This approach to urban water management is called ‘separate collection network’, as seen in the last section. However, many sewer systems do not have, yet, a WWTP (in fact, most sewage load does not yet undergo treatment in the country). Thus, when important storm water surges reach the sewers, they first affect low parts of their contribution areas. Sewage may return to premises within such areas (houses, commercial buildings, etc.) through sinks or drains, or leak upwards at inspection points (e.g., public network manholes). In addition, if excessive sewage flows into a treatment plant and exceeds its capacity, it may disturb the treatment process and hinder effluent quality – and that of the receiving waters (a river, channel, estuary, reservoir or sea). In Brazil, despite a clear choice for separate collection systems, these are facing growing difficulties of both the types mentioned above. Their main reasons are:

a) Fast and uncontrolled urbanisation may lead to improper refurbishment of private premises (often not supervised by an engineer), increasing the (illicit) trend to connect a building drainage pipe into its sewage system. This may be the case, for instance, in ancient areas where roads received too many upgrade layers on the top of previous ones, reaching too high levels. Thus, existing storm-water pipe outlets from premises may become lower than the road into which they discharge. In fact, it is common to observe outlets discharging at a too low position or, perhaps, the pipe stretch across a walking pavement be either ruptured or clogged. In fact, frequent public works (or even private ones, e.g., cable installation by telecommunication companies) may disregard the physical integrity of such pipes, either breaking or interrupting them. Figures 2 and 3 show some pictures randomly taken around a few blocks in downtown São Paulo City, such as to show examples of these problems. Since plastic became their most common material (PVC, usually), almost a standard, many of these rain water connections are so often destroyed and disregarded, that householders may opt for misdirecting their roof water to building sewage connections (still within the premises). Thus, it follows its way into the public sewers, outside the view of sanitation officers – being also common that unqualified personnel conduct such works. Only very recently (March 2014, to apply from April 18), the Brazilian Association of Technical Standards [18] issued a norm on building refurbishment so that it be designed and conducted

by an engineer or an architect – according to the nature of physical changes. The cases shown in Figure 3 illustrate the importance of this norm: a ramp for wheel chairs was being constructed at the cross point of a drainage pipe whose outlet at the gutter was visible. Nonetheless, the pipe was already ruptured when the first author saw the problem; required the presence of the engineer in charge, and suggested a more appropriate place for the ramp. Since the building company, hired by the Municipality, had no proposal of a position for replacing the pipe outlet, that suggestion was accepted and the pipe and the pavement were rebuilt at the original site.

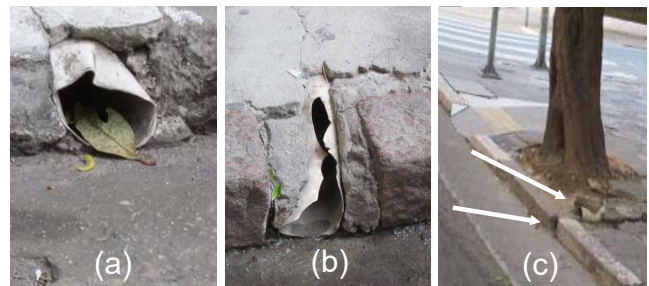


FIGURE. 2

OUTLETS INTO GUTTERS, EITHER POORLY KEPT (a; b; c: UPPER PIPE), OR AT A TOO LOW HEIGHT (c : LOWER PIPE): HIGIENÓPOLIS, SÃO PAULO (SP)



FIGURE. 3

DRAINAGE CONNECTORS DAMAGED BY THE INSTALLATION OF A RAMP FOR WHEEL CHAIRS: HIGIENÓPOLIS, SÃO PAULO (SP)

b) Poor maintenance of drainage systems may cause storm water leakages through the ground and infiltration into ageing, ruptured sewer pipes – thus, increasing their flow. This may happen either inside a building, from a pluvial pipe to a sewer, or from the public drainage to the public sewers.

Trying to alert Brazilian professionals to this important reality at fast growing cities and towns, the first author has been conducting field research on the occurrence of this phenomenon since 1999 [5] [6] [7] [8]. He also proposes a mathematical model [9] to forecast impacts of CSO events to water flow and quality in receiving rivers. It is true that the importance of this kind of system-operation tools will only become evident after the restoration of metropolitan river water quality in Brazil. However, the solution to this problem will be unreachable, at that point, if governments continue to be over lenient in what concerns to their legal obligation to fight illicit drainage-sewer connections, at all levels. A previous publication by the authors yields a comprehensive view of the problem, including modelling and control [19].

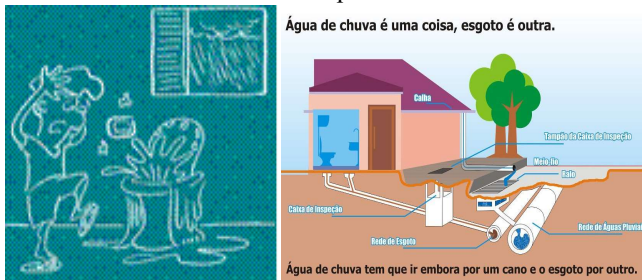
BRAZILIAN SANITARY AGENCIES FIGHT THE CSO WAVE THROUGH EDUCATION AND LEGISLATION

Brazilian sanitary authorities conscience of the danger of illicit, drainage-sewer connections during the sanitation boom (chiefly in São Paulo State, years 1970) was almost nil. This situation remained until about 1990, when the first warnings for the problem started – including the early publications by the first author on this issue. Thus, most of the collection and treatment system projects being built then were designed for dry weather flow, only. In the last few years, the consequences of CSOs to sewage collectors and treatment plants have been so astonishing that governments and sanitation agencies started to warn the population, and all sort of economies, for the need to reduce illicit connections. The following reports summarise this action.

São Paulo State Sanitation Agency, SABESP, started a statewide, on-line campaign after the call ‘storm water and sewage must have independent outlets’ [20]. Since Feb/2013, it also offered a complementary, on-line, illustrated leaflet to inform householders on capital queries on this subject.

The Ponte Nova Municipality, SP, started a campaign on the call ‘Storm water is a thing, sewage is another thing’ [21]. The scheme of Figure 4 alerts for this difference. Similar campaigns have been conducted by other municipalities: Araçatuba, SP: ‘Take care of storm water in the sewage’ [22], and Porto Ferreira, SP: ‘Sewers are neither the right place for solid waste, nor for storm water’ [23]. In Capivari, SP, the population is warned for possible backflows from sewers into contributing houses, in case of storm flows, due to illicit drainage-sewage pipe connections [24], and that new houses with such connections will not receive residence permits. Nova Odessa, SP, has had frequent sewer ruptures due to excessive flow during storms – and the Municipality has urged householders to disconnect illicit links [25].

The Municipality of Jaboticabal, SP, carries out the campaign ‘If your house storm water is going to the sewers, solve this problem before the worst happens’. On its on-line leaflet and appended files, they offer support to householders willing to regularise their connections [26]. Leme, SP, municipal campaign gives a step forward, stating that ‘Rainwater in the sewers may disturb the WWTP’ [27], and reports a plant inflow augmentation of 250% due to rain water incursion into sewers. They started another campaign: ‘Houses discharging storm water into sewers may be fined’ [28]. Since 31/5/2014, a relevant fine may apply to any house in discordance to the national separate-sewer standards.



Source: [20]

Source: [21]

FIGURE 3

ILLUSTRATIVE LEAFLETS ON CSO FROM SÃO PAULO STATE TOWNS

The Government of the State of Minas Gerais started he campaign ‘Take care of your house and your town: do not direct storm water into sewers and avoid trouble’. An on-line leaflet lists consequences of this misconduct, as well as necessary care to avoid them [29]. Another publication by COPASA (Minas Gerais State Sanitation Agency) [30] alerts for the penalty to apply to houses discharging stormwater with sewage: instant disconnection from the public network.

In the Center-West Brazilian States, investigations are being conducted on this problem, and campaigns carried out to wake public conscience up for the protection of sewers against irregular, storm water incursions – as it is the case of Goiás [31]. In Mato Grosso State, Tangará da Serra detected



FIGURE 4

WASTEWATER FLOOD: DOWNTOWN
TANGARÁ DA SERRA-MT, JAN/2014 [32]

CONCLUSIONS AND RECOMMENDATIONS

Conclusions and recommendations for the future arise from the discussions above, namely:

- 1) The escalating problem of stormwater incursions into Brazilian separate sewers and treatment plants, not designed for so high loads – and not to be designed so in the future – legally demands strong, effective actions by authorities [33], particularly in terms of educating designers to properly conceive sanitary facilities, and users to follow the basic rules that will keep systems integrity, and good performance.
- 2) Sectors of public administration must communicate and work harmoniously: municipalities; state governments; water industry; concession control agencies; technical professional boards. On legal grounds, authorities should enforce the law for users to follow rules and apply education programmes. Liable cases much on sight shall be addressed soon. Penalties must apply to whoever connects drainage to sewers.
- 3) Even if vast efforts start now to prevent and revert storm surges in Brazil, part of the problem will remain. And, given the amplitudes of surges, plant size expansion is not a feasible solution. Thus, dynamic control seems fundamental to solve it, necessarily associated to system monitoring, WWTP process modelling, river water quality simulation and, also, simulation of all system controls – thus, providing decision makers with a convenient tool for planning, design and real time operation. This might be allied to real time simulation of the hydraulic controls at collectors and fluvial structures – all based on a wastewater system monitoring framework [9].
- 4) After a ‘long winter hibernation’, the Brazilian sanitation community awoke for the reality that illicit drainage-sewer connections exist. Efforts to revert the problem and curb its

consequences are growing fast, representing ‘a light at the end of the tunnel’ for river water quality.

REFERENCES

- [1] Campos, C.M.: “Os rumos da cidade: urbanismo e modernização em São Paulo”. São Paulo: Editora SENAC, 2002.
- [2] Sevcenco, N.: “Orfeu extático na metrópole: São Paulo: sociedade e cultura nos frementes anos 20”. São Paulo: Cia. das Letras, 1992.
- [3] Telles, P.C.S.: “História da Engenharia no Brasil”. 2v. Rio de Janeiro: Clavero Ed., 1993.
- [4] Rodrigues, J.M.C., Jordão, E.P.: “Planning for wastewater treatment in São Paulo”. *Journal of the USA Pollution Control Federation*, Vol.61, No.7, 1989, pp.1190-97.
- [5] Reda, A.L.L., Ferreira, M.P.: “Esgoto combinado em rede separadora na RMSP: impacto de tormenta no afluente na ETE de Riacho Grande, São Bernardo do Campo”. Proceed.: 24. Congresso Brasileiro de Engenharia Sanitária e Ambiental. CD. R.de Janeiro: ABES, 2007.
- [6] Mello, G.S.L., Reda, A.L.L.: “Impacto de tormentas urbanas sobre a variação da qualidade do esgoto sujeito a inclusões de drenagem”. Proceed.: 22. Congresso Brasileiro de Engenharia Sanitária e Ambiental. CD. Rio de Janeiro: ABES, 2003.
- [7] Reda, A.L.L., Mello, G.S.L.: “Esgoto unitário ‘não-intencional’: futuro ameaçado por tormentas ou desafio para o planejador no futuro?”. Proceed.: 15. Simpósio Brasileiro de Recursos Hídricos. CD. Porto Alegre: ABRH, 2003.
- [8] Reda, A.L.L., Souza, T.G.L., Freitas, J.L.M.: “Impacto de chuva torrencial na qualidade do esgoto em rede separadora: perturbações no afluente à lagoa de estabilização de Lins, SP. Proceed.: 24. Congr. Brasil. de Eng. Sanitária e Ambiental. CD. ABES, R. de Janeiro, 2007.
- [9] Reda, A.L.L.: “Simulation and control of stormwater impacts on river water quality”. Tese (PhD). Imperial College of Science, Technology and Medicine, Londres, 1996, 512p.
- [10] Lessard, P., “Operational water quality management: control of stormwater discharges”. Tese (Ph.D.), Imperial College of Science, Technology and Medicine, Londres, 1989, 390p.
- [11] Beck, M.B.: “Operational water quality management: beyond planning and design”. Executive report 7, Laxenburg, Austria: IIASA, 84p., 1981.
- [12] Schuetze, M.R.: “Integrated Simulation and Optimum Control of the Urban Wastewater System”. Thesis (PhD) Imperial College of Science, Technology and Medicine, London, 1998, 450p.
- [13] Reda, A.L.L., Beck, M. B.: “Identification and Application of a Dynamic Model for Operational Management of Water Quality”. *Wat. Sc. Tech.* Vol.30, No.2, 1994, p.31-41.
- [14] Reda, A.L.L., Beck, M.B.: Ranking strategies for stormwater management under uncertainty: sensitivity analysis. *Wat.Sc.Tech.* Vol.36, No.5, 1997, p.357-71.
- [15] Reda, A.L.L., Beck, M.B.: “Simulation Model for Real-Time Decision Support in Controlling the Impacts of Storm Sewage Discharges”. *Wat. Sc. Tech.*, Vol.9, No.9, 1999, p.225-33.
- [16] Reda, A.L.L., Pamboukian S.V.D., Bonilha C.A., Kalil, S.: “Análise comparativa de sensibilidade na aplicação de simulação de qualidade fluvial visando a gestão dinâmica do tratamento de efluentes em duas regiões: tropical e fria”. Proceed.: X Safety, Health and Environment World Congress. CD. Washington, D.C.: IEEE, 2010.
- [17] Boksiner, G.; Jackson, D.; Terril, D.; Young M.; Anderson, D.: “Managing the deluge: the Trinity River Authority builds a flexible peak flow storage system with robust and repurposed preliminary treatment”. *Wat. Env. Tech.*, Vol.26, No.4, 2014, pp.28-33.
- [18] Associação Brasileira de Normas Técnicas. “NBR 16280:2014: reforma em edificações: sistema de gestão de reformas: requisitos”, Norm. 18/Mar/2014. Rio de Janeiro: ABNT.
- [19] Reda, A.L.L., Mendes, M., Ferreira, P.: “A new responsibility for the Brazilian water industry in the Era of Privatization: dynamic plant management for the treatment of storm and waste combined urban waters”. Proceed.: Adv. in Product. Manag. Systems: Competitive and Sustainable Manufacturing Products and Services. Cernobbio, Como, Italy, 11-13 Oct., 2010. Milan: Politecnico di Milano, 1.Vol., Pen drive.
- [20] Companhia de Saneamento Básico do Estado de São Paulo. “Água de chuva e esgotos devem ter saídas independentes”. *Notícias*, 22/2/2013. São Paulo: SABESP. Available on: <http://site.sabesp.com.br/site/imprensa/noticias-detalle.aspx?secaoId=65&id=5054>, 2013. Consulted on: 22/5/14.
- [21] Prefeitura Municipal de Ponte Nova. “Cuidados com a água de chuva na rede de esgoto”. *Notícias*, 13/6/2013. Ponte Nova, SP: Prefeitura Municipal de Ponte Nova. Available on: http://www.pontenova.mg.gov.br/home/index/principal/noticia.asp?i_d_texto=633869, 2013, 1p. Consulted on: 10/4/14.
- [22] Soluções Ambientais de Araçatuba. “Cuidado com a água de chuva no esgoto”. *Meio Ambiente*, 2014. Araçatuba, SP: SAMAR. Available on: <http://www.samar.eco.br/meio-ambiente/cuidado-com-a-agua-de-chuva-no-esgoto>, 2013. Consulted on: 23/5/14.
- [23] Prefeitura Municipal de Porto Ferreira. “Rede de esgoto não é lugar de lixo nem de água de chuva”. *Porto Ferreira Hoje*, 10/5/2012. Porto Ferreira, SP: PMPF. Available on: <http://www.portoferreirahoje.com.br/noticia/2012/05/10/rede-de-esgoto-nao-e-lugar-de-lixo-e-nem-de-agua-de-chuva/>, 2012. Consulted on: 20/5/14.
- [24] Serviço Autônomo de Água e Esgoto. “Água pluvial (água de chuva e de quintal)”. In: Os cuidados com a nossa água. Informative leaflet, 4p. Capivari, SP: SAAE, Prefeitura Municipal de Capivari. Available on: http://www.saaecapivari.com.br/site/wp-content/downloads/campanhas/cartilha_nossa_agua.pdf, Consulted on: 21/5/14.
- [25] Companhia de Desenvolvimento de Nova Odessa. “Chuva: CODEN faz apelo aos munícipes: rede não comporta volume de água”. *Novo Momento*: Cidades, 16/1/2014. Nova Odessa, SP: CODEN. Available on: <http://www.novomomento.com.br/Cidades/14112/chuva-coden-faz-apelo-aos-munipcipes>, 2014, 1p. Consulted on: 25/5/14.
- [26] Serviço Autônomo de Água e Esgoto de Jaboticabal, “Esgoto e água de chuva não se misturam!” Jaboticabal, SP: SAAEJ, Prefeitura Municipal de Jaboticabal. Available on: <http://www.saaej.sp.gov.br/novosite/noticias/87/esgoto-e-agua-de-chuva-nao-se-misturam>, 2014. 1p. Consulted on: 28/5/14.
- [27] Superintendência de Água e Esgotos da Cidade de Leme. “Água de chuva no esgoto pode comprometer ETE”. *Notícias*, June 2014. Leme, SP: SAECIL. Available on: <http://www.saecil.com.br/novo/index.php/noticias/91-agua-de-chuva-no-esgoto-pode-comprometer-ete>, 2014. Consulted on: 18/5/14.
- [28] Superintendência de Água e Esgotos da Cidade de Leme. “Residências que lançam água de chuva no esgoto poderão ser multadas”. *Notícias*, May 2014. Leme, SP: SAECIL. Available on: <http://www.saecil.com.br/novo/index.php/noticias/141-esgoto>, 2014, 1p. Consulted on: 20/5/14.
- [29] Companhia de Saneamento de Minas Gerais. “Água de chuva e esgoto não se misturam”. Belo Horizonte, MG: COPASA, Available on: <http://www.copasa.com.br/media2/Campanhas/AguaChuvaEsgotoNaoSeMisturam.pdf>, 2014. 2p. Consulted on: 20/5/14.
- [30] Agência Minas. “Ligações clandestinas da rede pluvial à rede de esgoto podem causar transtornos”. *Jornal Web*: Minas Gerais, 23/1/2013. Belo Horizonte, MG: Agência Minas. Available on: <http://triangulo-minas.blogspot.com.br/2013/01/ligacoes-clandestinas-da-rede-pluvial.html>, 2014. 2p. Consulted on: 10/4/14.
- [31] Saneamento de Goiás SA. “Água de chuva não é esgoto”. Goiânia, GO: SANEAGO. Leaflet, 1p. Available on: <https://www.saneago.com.br/relacionamento/wp-content/uploads/papelaria/AguaDeChuvaNaoEEsgoto.pdf>, not dated. Consulted on: 23/5/14.
- [32] Roberto, S. “Ligações indevidas de águas pluviais sobrecarregam rede coletora de esgoto”. *O Jornal*: Acontece MT, 29/1/2014. Tangará da Serra, MT. Available on: <http://www.ojornalmt.com/ligacoes-indevidas-de-aguas-pluviais-sobrecarregam-rede-coletora-de-esgoto/>, 2014. Consulted on: 8/3/14.
- [33] Governo Federal do Brasil. “Lei Nº 11.445, de 05/01/2007: Estabelece diretrizes nacionais para o saneamento básico. [Marco regulatório do saneamento básico]. Brasília: Presidência da República: Casa Civil. 2007. Available on: http://www.planalto.gov.br/ccivil_03/_ato2007-2010/2007/lei/111445.htm.